Team F

College0
Software Requirements Specification
For College0

Version 1.0

Revision History

Date	Version	Description	Author
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Software Requirements Specification

1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) is to provide an overview of the external behavior as well as the functional and nonfunctional requirements of College0: an AI-enabled online college program management system. The external behavior will be described in Section 2 via a use-case diagram, which depicts the variation in use cases depending on the actors. The functional and nonfunctional requirements will be given in Section 3 through reports for each use case followed by a supplementary section. Utilizing the outline of use-case modeling, this SRS will not only yield a sophisticated breakdown of College0, but also highlight the needs for such an application.

1.2 Scope

College0 aims to be a more superior interface compared to other online college login portals, such as CUNYFirst, which are plagued by issues such as login and timeout errors, poor design, dead links, and outdated or unclear information. It will greet all users with an appealing Graphical User Interface (GUI) displaying general information and provide customized experiences to each type of user: registrars, instructors, students, and visitors. These user types will be explained further in Section 1.3. College0's various features can be organized into eight subsystems: application, class set-up, course registration, class running, grading, reviews, complaints, and Q & A. The use cases within each of these subsystems depend on the type of user, and, as such, will be elaborated on in Sections 2 and 3, where we reference our use-case model. Through an engaging frontend, seamless navigation, the employment of AI, and other additional functionalities, College0 will be an especially user-friendly and feature-rich college program management system, allowing students to achieve their goals with ease and focus on their education.

1.3 Definitions, Acronyms, and Abbreviations

Registrar: An administrative official responsible for admitting or rejecting student and instructor applicants, setting up courses and their respective schedules, regulating reviews, supervising student and instructor performance, managing student graduation, and processing complaints. This user type has no restrictions and can therefore do anything on College0.

Instructor: A faculty member responsible for teaching courses, managing their course waitlists, and assigning grades. They can access courses they are assigned by registrars.

Student: A person enrolled at the college that can register for courses, write reviews, and apply for graduation.

Class Set-up Period: The first of four periods within a semester. During this period, registrars set up class times, course instructors, and class sizes.

Course Registration Period: The second period within a semester during which all matriculated students can enroll in two to four courses as long as there is no time conflict and the upper limit of the course is not reached.

Class Running Period: The third period within a semester during which classes are in session and students can no longer register for courses (except those who are allowed a special registration period). Students and instructors may be warned, classes may be cancelled, and instructors may be suspended.

Special Registration Period: A period within the class running period. Students who were enrolled in classes that were cancelled are given another chance to register for other courses.

Grading Period: The last of the four periods within a semester. During this period, instructors are expected to assign grades to students in their classes.

Taboo Word: A word deemed by a registrar to be prohibited from use in a review.

Honor Roll: A special distinction given to students with an exceptional GPA. One honor can be used to remove a warning.

Warning: A form of disciplinary action that can be issued to students and instructors. Students who have received up to three warnings will be suspended for one semester and must pay a fine to the registrar. Instructors that have accumulated three warnings will be suspended.

AI Chatbot: An Artificially Intelligent virtual assistant that provides local information about the college for users with questions about the system. Visitors can ask general questions about classes and requirements, students can ask questions on the classes they are taking, and instructors can ask questions about the students in their class(es).

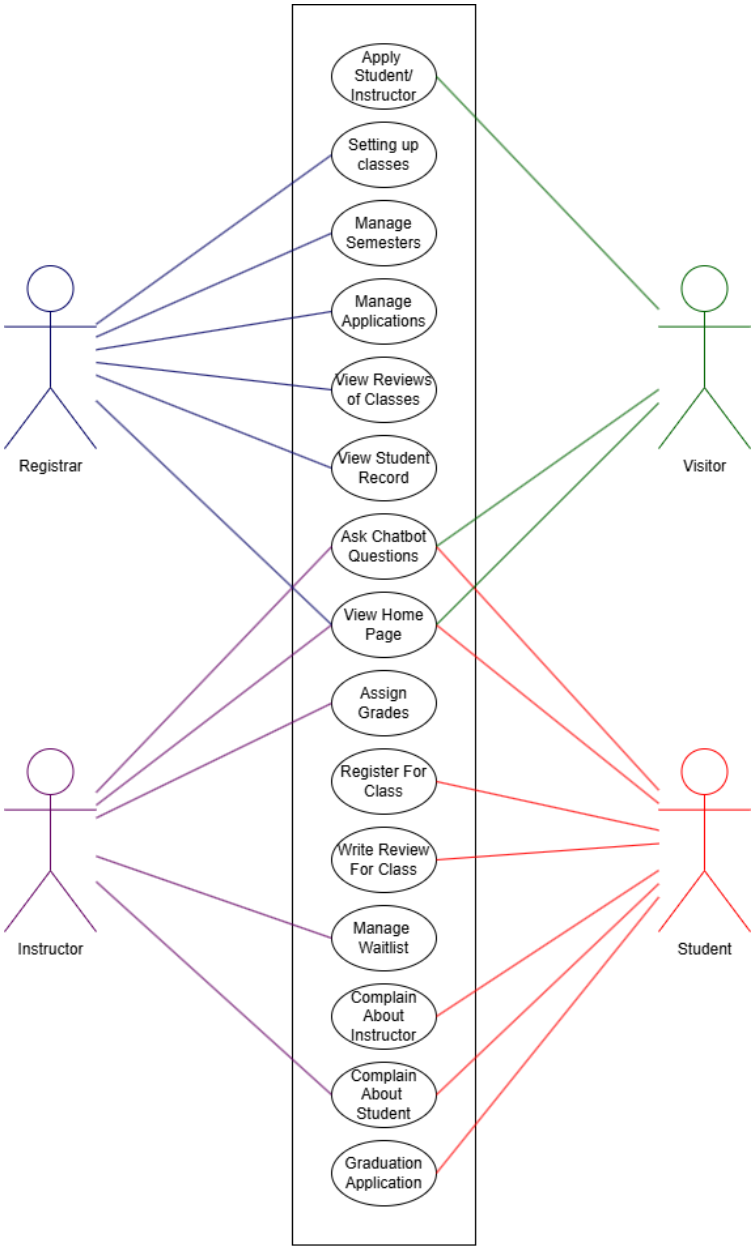
LLM: Stands for Large Language Model. A machine learning model designed for natural language processing and, in this case, generates text to mimic human responses. If a user's question to the AI chatbot cannot be answered, it will be directed to the LLM for a possible answer.

1.4 Overview

The rest of this SRS will provide additional context as well as describe the specific functionalities of College0. The context will be provided in Section 2: Overall Description, starting with a use-case diagram in Section 2.1. In the following Section 2.2, assumptions and dependencies needed for College0 to be a viable application will be detailed. Section 3: Specific Requirements will describe the various features of College0 based on the use-case diagram from Section 2.1. This will be achieved by laying out in-depth explanations of each use case in Section 3.1, followed by a description of additional functionalities not covered by the use cases in Section 3.2.

2. Overall Description

2.1 Use-Case Model Survey



A-1

Actors: All actors can view the home page which contains highest rated classes, lowest rated classes, and students with highest GPA.

- Visitors: Can apply to be a student or instructor and ask AI chatbot general questions about classes and requirements.
- Students: Can ask AI chatbot about classes they are taking, register for classes, write reviews for classes, apply for graduation, and file complaints about another student or instructor.
- Instructors: Can ask AI chatbot about students in classes they are teaching, assign grades, file complaints about students, and manage waitlists.
- Registrars: Can set up classes, view student records/reviews, and manage semesters/applications.

2.2 Assumptions and Dependencies

1. **Authentication**

   To ensure data privacy as well as proper authorization for each level of user College0 will use password, id, and email verification to ensure security. This enables College0 to ensure proper permissions are given to the proper users as well as prevent security breaches in the site.

2. **Data Storage**

   College0 will rely on persistent data storage to hold credentials, permissions, classes, student records, and reviews. The database will be depended on to access the sites, update information, and be able to securely store personal information. In addition the database will need to store taboo word lists and be able to be accessed to filter reviews.

3. **GUI**

   College0 will depend on a single GUI window to ensure seamless user experience. The GUI will contain a GPA leaderboard, as well as classes that have the highest and lowest ratings. In addition the GUI must be user friendly enabling ease of use, and real time updates with program data.

4. **Registrar Management System**

   The platform will rely on the registrars to be able to use the registrar management system in order to correctly set up the semesters and classes and manage reviews and complaints. The system will also rely on the registrars to create and manage the taboo word list as well as process visitor applications.

5. **AI Knowledge Base**

   The AI chatbot will depend on human ability to use AI chatbots and the local information store / LLM. The first dependency depends on the user understanding the risk of hallucinations and the ability to ask questions to get the desired information. The second dependency depends on the local information store to give basic information or the LLM to give a response if the local information store is unable to.

6. **Automation**

   College0 will depend on a background automated system that will moderate reviews with the taboo word list, automated reviews if student gpa drops below threshold or above, enrollment management, and faculty oversight. These will all be handled by an automated system that the registrar will have access to.

7. **Creative feature (College Swap)**

   College0 will stand out with our creative feature of College Swap. This feature will assume that the requirements will be the same for graduation and GPA requirements. It will also depend on the database to ensure that data between the colleges don’t get conflated in storage while still being accessible to registrars.

3. Specific Requirements

3.1 Use-Case Reports

<u>Application for Student/Instructor Enrollment</u>

The system offers an interface for allowing people visiting the website to enroll in the college, as a student giving them a user account with all the features associated with being a student, as well as allowing visitors to enroll as a professor, giving them access to the tools and features needed to manage a given class.

<u>Manage Applications</u>

Here the system must allow the registrars to receive applications for user accounts specifically, enrollment management for people applying to the college as a student, and faculty management for those applying to be instructors in a given department.

<u>View Home Page</u>

The system shall provide all users access to the home page of the college website, however depending on the permission level of the user account the actual content will be changed dynamically, based on the necessary features that they need to access, i.e. students being able to access their course grades and overall transcript, etc.

<u>Academic Semester/ Course management</u>

Manage Semesters

The system will allow registrars to set up/establish a new academic year, automatically creating a fall and spring semester for the next academic year starting in the Fall. It will have a clear and understandable user interface to prevent clerical errors.

Setting Up Classes

The system will provide registrars with the ability to produce classes with course codes, names, session times, as well as start and end dates for the current semester as well as allowing them to specify who will be teaching the course.

View Reviews of Classes/ Reports of students or professors

The system will allow the registrars to view how well any given class is doing, review-wise. More specifically, it will allow them to view what review students have put in over the course as well as viewing any reports made by students and professors about other students/professors. Furthermore, it provides registrars with the ability to notify professors if they have to explain the unsatisfactory reviews to the registrar.

View Student Record

The system will also allow the registrars to view the records of every single enrolled student at the college, providing them the ability to notify a student when they must schedule a meeting to discuss unsatisfactory grades.

Complaints, Students and Professors

Student Complaints

Both students and professors will have the ability to write complaints about students within any given class, this will happen during the course duration period creating an avenue for registrars to view dissatisfaction about a given student and allowing them to take further action if deemed necessary.

Professor Complaints

Students will be given the ability to report a professor teaching a given course, with a provided text area to describe the nature of the report, it functions essentially similar to the student complaints, however only students can report professors.

Classes

Class Registration

The system will allow students to register for classes during the class registration period, allowing students to register for up to 4 classes, as long as none of them have a time conflict with each other.

Waitlist Registration / Management

In the event that a given class is full during course registration, the students will have the option to apply for the class' waitlist, notifying the professor that a student has applied to register as a part of the waitlist

Moreover, the Professor will be notified when a student signs up for the waitlist as well as when a spot in the class is open to allow the student to be matriculated once the semester proper begins.

Grade Assignment

The system offers functionality that is built for the professors to be able to enter in student grades during the grading period of the semester, with warnings being given out to those professors who fail to meet the grading deadline.

General System Functionality

ChatBot

The system will allow users to ask a chatbot questions about general functionality of the system as well as how to navigate through the various features that the system offers and allows students

to ask what classes they still need and what classes are recommended to take, professors can also ask questions about where to find information on the class.

Graduation Application

The system will give students the ability to apply for graduation, this can happen at any point however if the student does not meet the requirements for graduation they will be warned, if they do meet the requirements they will be awarded with a bachelor's degree and automatically deletes the user's account.

Transfer Application

The system will give students the ability to apply for transfer, by giving them a school option to apply for, and giving them the same questions that most applicants to that college would need to answer in order to be considered for application.

3.2 Supplementary Requirements

Automated Termination System

The system handles things such as warnings and low GPA with an automated feature, if a student receives too many warnings, or has far too low of a GPA (< 2.0) they will be automatically terminated from the college.

It will also do something similar for professors that receive far to0 many low reviews on many of their classes, if the professor fails to report to the registrar the professor in question will also be automatically terminated from the college.

System Security

The system will make sure to handle all user actions with the proper level of permissions that the user should have. For example, students will only be able to view their own grades for any given class, whereas professors will be able to view grades for the classes that they are in charge of.

Data Storage

The system will have stable structured data storage, and will prevent accidental deletion by preventing the system from *truly* deleting any given information, instead setting any kind of data as "inactive" or "disabled" in order to make certain that if any information needs to be retrieved it is always accessible

User Interface

The system will have an easy to navigate, effective user interface allowing users to quickly understand what part of the site they are on, and how to navigate to specific parts of the website, as well as a small button that will bring up the AI chatbot on any page that the user is located.

System Stability

The system will be built as robustly as possible in order to prevent downtime, or at the very least minimize it, it will be tested rigorously in order to handle higher traffic during times of registration, as well as many users using the same feature at the same time.

4. Supporting Information

Appendices:

Appendix A: Use Case Diagram

Figure A-1 shows the different types of users on the left and right side with lines connecting them to what use cases they have access to.

Appendix B: CollegeSwap feature Logo Mockup

COLLEGE
SWAP™